ADVANCED EXCEL · LESSON 15 OF 16

Office Scripts, Power Automate and AI-Assisted Excel Formulas

Automate Excel on the web with Office Scripts written in TypeScript, run them on a schedule with Power Automate, and use AI assistants to draft formulas, with a checklist for testing every formula before you rely on it.

Gap-fill lesson | Excel for Microsoft 365, 2026 | Advanced | 120 min

By the end of this lesson you can

- Explain how Office Scripts differ from VBA macros
- Record and edit an Office Script in Excel on the web
- Trigger a script from Power Automate on a schedule
- Test an AI-written formula on edge cases before using it

Datasets used

`superstore_sales.csv`

Office Scripts: automation for Excel on the web

Office Scripts are automations written in **TypeScript** that run in Excel on the web and in Microsoft 365. They do not need a desktop macro, so they work on any device with a browser, and they can be started by Power Automate. Use them when the file lives in SharePoint or OneDrive and the team works in the browser.

	VBA macro	Office Script
Language	VBA	TypeScript
Runs in	Excel desktop	Excel on the web and desktop (Microsoft 365)
Started by	Button, shortcut, Alt+F8	Button, Power Automate, Excel on the web
File type	.xlsm	Normal .xlsx
Good for	Desktop-only tasks	Shared team files and scheduled runs

CASE STUDY **A Monday summary for the team**

A team lead wants a summary sheet rebuilt every Monday from the shared sales file: totals by region, a sort by sales and a timestamp. The file is in SharePoint and edited by five people. A script runs at 7 a.m. through Power Automate, so the summary is ready before the team arrives.

Step 1: record and open a script

1. Open the workbook in Excel on the web.
2. Go to **Automate > Record Actions**.
3. Format a table, add a total row and sort it. Click **Stop**.
4. Open **Automate > Code Editor** to see the script.

The recorded script uses the Office Scripts API. It is written as a function, with the workbook passed in as `workbook`:

TYPESCRIPT

```
function main(workbook: ExcelScript.Workbook) {
    let sheet = workbook.getActiveWorksheet();
    let range = sheet.getRange("A1:F1");
    range.getFormat().getFont().setBold(true);
    range.getFormat().getFill().setColor("#1E8E5A");
}
```

As with VBA, recorded code uses fixed addresses. Replace them with values that come from the data.

Step 2: make the script adapt

This version finds the used range and totals the sales column, whatever its size:

TYPESCRIPT

```
function main(workbook: ExcelScript.Workbook) {
    const sheet = workbook.getActiveWorksheet();
    const used = sheet.getUsedRange();
    const rowCount = used.getRowCount();

    // Column E holds sales; rows 2 to the last data row
    const sales = sheet.getRange(`E2:E${rowCount}`);
    const total = sales.getValues().reduce((sum, row) => sum + Number(row[0] || 0), 0);

    // Write the total in a labelled cell below the data
    sheet.getRange(`D${rowCount + 2}:E${rowCount + 2}`).setValues([["Total sales",
total]]);
    sheet.getRange(`E${rowCount + 2}`).getFormat().setColumnWidth(110);
}
```

The `reduce` call adds up the numbers in JavaScript. Writing a total as a value is fine for a report; if the total must update as data changes, write a formula with `setFormula("=SUM(E2:E...)")` instead.

::::tip Prefer formulas for live totals A script that writes a value stops being correct the moment the data changes. A script that writes a formula stays correct. Use values for a snapshot (for example, a Monday report that must not change later) and formulas for anything that people will keep editing. ::::

Step 3: run it on a schedule with Power Automate

Office Scripts can be called from a Power Automate flow:

1. Create a flow with the trigger **Recurrence**, set to every Monday at 7:00.
2. Add the action **Excel Online (Business) > Run script**.
3. Choose the workbook from SharePoint or OneDrive, and the script name.
4. Optional: add **Send an email** to notify the team after the run.

Test the flow once by running it manually, and open the workbook to check the result before you rely on the schedule.

::::warn Scheduled runs need care A scheduled script runs without anyone watching. Make sure it cannot overwrite the source data, works on a copy if the file is shared, and sends a notice when it fails. Keep the raw data on a separate sheet from the output. ::::

AI-assisted formulas

AI tools inside Excel, such as Copilot, and general AI assistants can write formulas from a plain-language request. This is useful for drafting, and dangerous when the result is trusted without checking.

A good request names the columns, the logic and the output:

> "In column F, show 'High' if sales in column E are above 5000 and the region in column C is 'North' or 'South', otherwise 'Standard'. Use one formula."

A reasonable answer is:

```
=IF(AND(E2>5000, OR(C2="North", C2="South")), "High", "Standard")
```

Read the formula before you use it. Ask: does it match the rule, the column letters and the case of the text?

Testing an AI-written formula

Every formula from an AI tool should pass four tests before it enters a report:

1. **Known answer**: pick three rows where you know the right result by hand and check them.
2. **Boundary values**: test exactly 5000 (is it "above"?), blank cells and zeros.
3. **Unexpected text**: test "north" in lower case, extra spaces and misspellings.
4. **Totals**: compare the total of the new column with an independent calculation, such as a pivot table.

Test cases for the High/Standard formula

	A	B	C	D	E
1	Sales	Region	Expected	Formula result	Pass?
2	6000	North	High	High	yes
3	5000	North	Standard	Standard	yes
4	7000	north	High	High	yes

```
D2    =IF(AND(A2>5000,OR(B2="North",B2="South")),"High","Standard")
E2    =IF(D2=C2,"yes","NO")
D3    =IF(AND(A3>5000,OR(B3="North",B3="South")),"High","Standard")
E3    =IF(D3=C3,"yes","NO")
```

```
D4    =IF(AND(A4>5000,OR(B4="North",B4="South")),"High","Standard")
E4    =IF(D4=C4,"yes","NO")
```

All three rows pass. The third row passes because Excel's text comparison (=) ignores case, so "north" matches "North". That is the behaviour you want here, but you should know it is a choice, not an accident. If a future rule needed case-sensitive matching, the test would catch the difference only if you had written a case-sensitive expectation.

::::note Case sensitivity in Excel Excel comparisons with = ignore case, so "north" equals "North". Use `EXACT(a, b)` only when case must matter. ::::

Choosing the right tool

- **Repeated steps on a shared file in the browser**: Office Scripts, run by hand or by Power Automate.
- **Repeated steps on a desktop file**: a VBA macro.
- **Combining and cleaning many files**: Power Query, which refreshes with one click.
- **A one-off formula**: write it yourself, or check an AI draft with the four tests.

::::key What to remember - Office Scripts are TypeScript automations for Excel on the web; they run without desktop macros. - Power Automate can run a script on a schedule; test the flow manually first. - Prefer formulas over values for totals that must keep updating. - Test every AI-written formula with a known answer, boundary values, unexpected text and a total check. ::::

Quick quiz

5 questions · tap an answer to check it

1. What language are Office Scripts written in?

A VBA

B Python

C TypeScript

D C#

Answer: C. Office Scripts use TypeScript and the Office Scripts API.

2. Which tool can run an Office Script every Monday without someone opening the file?

A Power Query

B Power Automate with a Recurrence trigger

C Goal Seek

D Conditional Formatting

Answer: B. A Recurrence trigger starts the flow on a schedule, and the flow runs the script.

3. Why should a script that writes a total use a formula rather than a value if people keep editing the data?

A Values cannot be stored

B The value stops being correct when the data changes, while a formula updates

C Formulas are faster to write

D Scripts cannot write values

Answer: B. Formulas recalculate; a pasted value does not.

4. An AI-written formula gives the right result for the rows you tested. What should you do next?

A Use it in the report immediately

B Test boundary values, blanks and unexpected text before relying on it

C Delete it

D Convert it to a macro

Answer: B. Passing one set of rows does not prove the formula handles every case.

5. Does = in Excel distinguish "North" from "north"?

- **A** No, text comparison with = ignores case
- **B** Yes, always
- **C** Only in pivot tables
- **D** Only on Excel for the web

Answer: A. Use EXACT if case must matter.

EXERCISE **A scheduled summary**

Record an Office Script that takes a sheet of sales lines, creates a new sheet called "Summary" with total sales by region using formulas (`=SUMIF`), and writes the run time in a labelled cell. Then write the Power Automate flow that runs it every weekday at 8:00. List the four tests you would run on the summary before the team sees it.

Show solution (worked answer)

Hints: in the script, use `workbook.addWorksheet("Summary")` and `setFormula` with `=SUMIF(Data!C:C, A2, Data!E:E)`. For the timestamp, write `=NOW()` or the script's `new Date()` value with a label. Test with a known total, a region that has no sales, a blank row in the data, and a changed source value to see whether the summary updates.

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